SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code: CSA10 / CSA1005	Course Title: Software Engineering
CO Assessed: CO1	Assessment: Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage: 20%	Total Marks: 25
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Date: _____	Duration: 60 minutes

CO1 Statement: Apply advanced methodologies and agile project management techniques to manage software projects effectively.

Code of Conduct

I A Mohammad Naheed /192525336 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Mohammad Naheed

Annexure A – Agile Sprint Planning & User Story Workshop

Problem Statement No. 33: Smart Courier and Last-Mile Delivery Management Platform

1. Identify the Project Vision and Stakeholders

1.1 Project Vision

Vision Statement: "For logistics and e-commerce companies that struggle with failed deliveries, inefficient routes and poor customer visibility, the Smart Courier and Last-Mile Delivery Management Platform is a cloud-based system that automatically assigns shipments, plans optimal routes, tracks every delivery attempt and keeps customers informed in real time. Unlike manual or spreadsheet-based dispatching, our platform reduces delivery cost and delay while improving customer satisfaction."

1.2 Project Objectives

- Manage shipment information, customer addresses and delivery schedules in one central system.
- Assign deliveries to agents using location, priority, vehicle capacity and operational constraints.
- Plan efficient delivery routes and track delivery status in real time.
- Record every delivery attempt and manage failed deliveries with rescheduling or return workflows.
- Notify customers at every important event and generate delivery analytics.

- Achieve the expected outcomes: improved delivery efficiency, fewer failed attempts, better route utilisation, better customer visibility, fewer delays and lower operational cost.

1.3 Key Stakeholders and Their Roles

Stakeholder	Role in the Project	Interest / Expectation
Product Owner (Logistics Business Head)	Owns the product backlog, sets priorities and accepts completed work.	Maximum business value, lower cost per delivery.
Scrum Master	Facilitates Scrum events, removes impediments, coaches the team.	Smooth, predictable sprint flow.
Development Team (developers, UI/UX designer, QA tester)	Design, build, test and deliver the increment each sprint.	Clear stories, realistic sprint commitment.
Delivery Agents	End users of the mobile app; deliver parcels and update status.	Simple app, fair workload, efficient routes.
Dispatcher / Hub Manager	Manages assignments, monitors delivery progress, handles exceptions.	Automatic assignment, live control dashboard.
Customers / Receivers	Provide addresses, receive parcels and notifications.	Accurate tracking, on-time delivery, easy rescheduling.
Customer Support Team	Handles queries, complaints and rescheduling requests.	Fewer "where is my parcel" calls.
E-commerce Sellers / Partners	Send shipment orders and receive status and returns data.	Reliable delivery promise, fewer returns.
Operations / Finance Manager	Uses analytics to control cost and plan capacity.	Accurate reports and cost visibility.
System Administrator / IT Team	Deploys, secures and maintains the platform.	Security, uptime and easy maintenance.

2. Create Epics and User Stories

2.1 Epics

Epic ID	Epic Name	Description
E1	User and Access Management	Secure login and role-based access for admin, dispatcher, agent, support and customer.
E2	Shipment Management	Create, import, update, search and cancel shipments with priority and delivery window.
E3	Address Management	Capture, validate, geocode and store customer addresses and delivery preferences.

Epic ID	Epic Name	Description
E4	Agent and Vehicle Management	Maintain agent profiles, availability, zones, vehicles and capacities.
E5	Delivery Assignment and Route Planning	Assign shipments to agents under constraints and generate optimised routes.
E6	Delivery Tracking and Attempts	Status life-cycle, delivery attempt recording and proof of delivery.
E7	Failed Delivery Management	Failure reasons, rescheduling, retry limits and return-to-sender.
E8	Customer Notification and Visibility	SMS, e-mail and push alerts and a customer tracking page.
E9	Analytics and Reporting	Dashboards and reports on delivery performance and cost.

2.2 User Stories

Format used: As a <user>, I want <feature>, so that <benefit>.

Story ID	Epic	User Story
US-01	E1	As a system user, I want to log in with my role-based account, so that I can access only the features meant for me.
US-02	E2	As a dispatcher, I want to create and edit shipments with weight, size, priority and delivery window, so that all parcel details are stored correctly.
US-03	E2	As a seller, I want to upload shipments in bulk or through an API, so that I can send many orders without manual entry.
US-04	E3	As a customer, I want to enter and confirm my delivery address, so that the agent can find my location without confusion.
US-05	E3	As a customer, I want to save a preferred delivery time slot and landmark, so that I am available when the parcel arrives.
US-06	E4	As a hub manager, I want to register agents and their vehicles with capacity and shift timings, so that assignments respect real limits.
US-07	E5	As a dispatcher, I want shipments to be assigned automatically to suitable agents, so that I save planning time and balance workload.
US-08	E5	As a delivery agent, I want an optimised route with stop sequence, so that I travel less and deliver on time.
US-09	E6	As a delivery agent, I want to update delivery status from my mobile app, so that everyone sees the latest position of the parcel.
US-10	E6	As a delivery agent, I want to record each attempt with time, location and proof of delivery, so that there is a reliable delivery record.

Story ID	Epic	User Story
US-11	E7	As a delivery agent, I want to select a reason when a delivery fails, so that the system can take the right follow-up action.
US-12	E7	As a customer, I want to reschedule a failed delivery or change the address, so that I can still receive my parcel.
US-13	E8	As a customer, I want to track my parcel on a tracking page, so that I know where it is and when it will arrive.
US-14	E8	As a customer, I want SMS and e-mail alerts for dispatch, arrival and failure, so that I do not need to call support.
US-15	E9	As an operations manager, I want a dashboard of on-time rate, first-attempt success and cost per delivery, so that I can improve performance.
US-16	E5	As a dispatcher, I want dynamic re-routing when an urgent shipment or cancellation occurs, so that routes stay efficient during the day.

3. Define Acceptance Criteria

Acceptance criteria are written in Given–When–Then format for the user stories selected for Sprint 1 (US-01, US-02, US-04, US-06, US-07, US-09, US-13).

US-01: Role-based login

No.	Acceptance Criteria (Given – When – Then)
AC1	Given a registered user with valid credentials, When the user logs in, Then the system opens the dashboard for the user's role within 3 seconds.
AC2	Given a wrong password, When the user submits it three times, Then the account is locked for 15 minutes and an error message is shown.
AC3	Given a delivery agent is logged in, When the agent tries to open the admin page, Then access is denied.

US-02: Create shipment

No.	Acceptance Criteria (Given – When – Then)
AC1	Given a dispatcher on the new-shipment form, When all mandatory fields (sender, receiver, address, weight, priority) are filled and saved, Then a unique shipment ID is generated and the status is "Created".
AC2	Given a mandatory field is empty, When the dispatcher clicks Save, Then the system blocks saving and highlights the missing field.
AC3	Given a saved shipment, When the dispatcher searches by shipment ID, Then the record is displayed in less than 2 seconds.

US-04: Address entry and validation

No.	Acceptance Criteria (Given – When – Then)
AC1	Given a customer enters an address, When the address is submitted, Then the system standardises the format and finds latitude and longitude coordinates.
AC2	Given an address that cannot be located, When it is submitted, Then the system asks the user to correct it or drop a pin on the map.
AC3	Given a customer with an existing address, When the same address is entered again, Then the system warns about a duplicate.

US-06: Register agents and vehicles

No.	Acceptance Criteria (Given – When – Then)
AC1	Given a hub manager on the agent form, When name, phone, zone, shift, vehicle type and capacity are saved, Then the agent becomes available for assignment.
AC2	Given an agent marked "On Leave", When assignment runs, Then no shipment is assigned to that agent.
AC3	Given a vehicle capacity value less than or equal to zero, When the form is saved, Then the system rejects it with an error message.

US-07: Automatic assignment

No.	Acceptance Criteria (Given – When – Then)
AC1	Given unassigned shipments and available agents, When the dispatcher runs assignment, Then every shipment is assigned to an agent in the same delivery zone.
AC2	Given an agent's vehicle capacity, When shipments are assigned, Then the total weight and volume of assigned parcels never exceed that capacity.
AC3	Given urgent and normal shipments, When assignment runs, Then urgent shipments are allocated first, and the dispatcher can manually override any assignment.

US-09: Agent updates status

No.	Acceptance Criteria (Given – When – Then)
AC1	Given an agent has assigned shipments, When the agent taps "Out for Delivery" or "Delivered", Then the status changes and a time stamp is stored.
AC2	Given the agent's phone has no network, When a status is updated, Then it is saved on the device and synchronised automatically within 1 minute of reconnecting.
AC3	Given a status change is saved, When the dispatcher refreshes the dashboard, Then the new status is visible within 5 seconds.

US-13: Customer tracking page

No.	Acceptance Criteria (Given – When – Then)
AC1	Given a valid tracking number, When the customer opens the tracking page, Then the current status and status history are shown.

No.	Acceptance Criteria (Given – When – Then)
AC2	Given an invalid tracking number, When it is entered, Then a clear "not found" message is displayed.
AC3	Given a small mobile screen, When the page is opened, Then it displays correctly without horizontal scrolling.

4. Prioritize the Product Backlog

The product backlog is ordered using business value, customer need and technical dependency, and classified with the MoSCoW method (Must have, Should have, Could have). Stories that other stories depend on, such as login, shipment, address and agent registration, are placed first.

Rank	User Story (Short Title)	Business Value	MoSCoW	Depends On	Planned Sprint
1	US-01 Role-based login	High	Must	None	Sprint 1
2	US-02 Create shipment	High	Must	US-01	Sprint 1
3	US-04 Address entry & validation	High	Must	US-01	Sprint 1
4	US-06 Register agents & vehicles	High	Must	US-01	Sprint 1
5	US-07 Automatic assignment	Very High	Must	US-02, US-04, US-06	Sprint 1
6	US-09 Agent status update	High	Must	US-07	Sprint 1
7	US-13 Customer tracking page	High	Must	US-09	Sprint 1
8	US-08 Optimised route	Very High	Must	US-07	Sprint 2
9	US-10 Record attempt & proof	High	Must	US-09	Sprint 2
10	US-11 Failure reason	High	Must	US-10	Sprint 2
11	US-14 SMS / e-mail alerts	High	Should	US-09	Sprint 2
12	US-12 Reschedule failed delivery	Medium	Should	US-11, US-14	Sprint 3
13	US-05 Preferred slot & landmark	Medium	Should	US-04	Sprint 3
14	US-03 Bulk / API shipment upload	Medium	Should	US-02	Sprint 3
15	US-15 Analytics dashboard	Medium	Should	US-10, US-11	Sprint 3

Rank	User Story (Short Title)	Business Value	MoSCoW	Depends On	Planned Sprint
16	US-16 Dynamic re-routing	Medium	Could	US-08	Sprint 4

5. Plan the Sprint Backlog

5.1 Sprint Details

Item	Details
Sprint Number / Length	Sprint 1 / 2 weeks (10 working days)
Scrum Team	1 Product Owner, 1 Scrum Master, 3 Developers, 1 UI/UX Designer, 1 QA Tester
Team Capacity (velocity assumption)	About 32 story points per sprint, based on 6 working team members × 10 days with 20% allowance for meetings and leave
Selected Stories	US-01, US-02, US-04, US-06, US-07, US-09, US-13 (total 31 points)

5.2 Sprint Backlog with Task Breakdown

User Story	Development Tasks	Estimated Effort
US-01 Role-based login	T1: Design user and role tables T2: Build login API with password hashing T3: Implement role-based access checks T4: Create login screens (web and mobile) T5: Write unit and security tests	14 h
US-02 Create shipment	T1: Design shipment table and status codes T2: Build create, edit, search APIs T3: Create shipment form and list screens T4: Add field validation and unique ID logic T5: Test and fix defects	26 h
US-04 Address validation	T1: Design address table T2: Integrate map / geocoding API T3: Build address form with map pin T4: Add standardisation and duplicate check T5: Test with sample addresses	26 h
US-06 Register agents & vehicles	T1: Design agent and vehicle tables T2: Build agent and vehicle APIs T3: Create registration and availability screens T4: Test validations	16 h

User Story	Development Tasks	Estimated Effort
US-07 Automatic assignment	T1: Define assignment rules (zone, priority, capacity) T2: Implement assignment algorithm service T3: Build manual override screen T4: Create assignment result view T5: Unit and scenario tests with capacity limits T6: Review and refactor	40 h
US-09 Agent status update	T1: Design status event table T2: Build status update API T3: Create agent app task list and update buttons T4: Implement offline storage and sync T5: Test with network loss	26 h
US-13 Customer tracking page	T1: Build tracking-number lookup API T2: Create responsive tracking page T3: Show status history T4: Test on mobile and desktop	14 h

Common tasks for the sprint: set up the project repository, development environment and CI pipeline (8 h); prepare the test data and deployment to the test server (8 h); integration testing and sprint demo preparation (10 h).

6. Estimate Effort Using Story Points

6.1 Estimation Technique

The team used Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13). Each member privately selected a card based on complexity, amount of work and uncertainty. When estimates differed widely, the highest and lowest estimators explained their reasons and the team voted again until agreement. A small, well-understood story, US-01 being the reference, was taken as a baseline of 2 points.

6.2 Story Point Estimates and Justification

User Story	Points	Justification
US-01 Role-based login	2	Well-known pattern with standard libraries; low complexity and low risk.
US-02 Create shipment	5	Multiple screens, validation rules, database design and search; moderate effort but familiar work.
US-04 Address entry & validation	5	Depends on an external map API, address standardisation and duplicate detection; some uncertainty.
US-06 Register agents & vehicles	3	Simple data management with few business rules.
US-07 Automatic assignment	8	Most complex story: algorithm with zone, priority and capacity constraints, override option and heavy testing; highest risk.

User Story	Points	Justification
US-09 Agent status update	5	Mobile app work plus offline storage and synchronisation, which adds complexity.
US-13 Customer tracking page	3	Read-only page with an API and responsive design; low risk once status data exists.
Total	31	Within the estimated team capacity of 32 points, leaving a small buffer.

7. Present the Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, a dispatcher can log in, create shipments with validated customer addresses, assign them automatically to available agents within capacity limits, and delivery agents can update the delivery status so that customers can track their parcels online."

7.2 Expected Sprint Deliverables

- Secure, role-based login for admin, dispatcher, agent and customer.
- Shipment management module (create, edit, search) with unique shipment IDs.
- Address management with map-based validation and geocoding.
- Agent and vehicle registration with availability and capacity.
- Automatic assignment engine with manual override.
- Agent mobile app with task list, status update and offline synchronisation.
- Customer tracking page with status history.
- Working, tested increment deployed on the test server, with sprint demo, updated product backlog and burndown chart.

7.3 Definition of Done

- Code is reviewed and merged into the main branch.
- Unit and integration tests pass and all acceptance criteria are met.
- No open critical or high-priority defects.
- Feature is deployed to the test environment and accepted by the Product Owner.
- Documentation (API notes and user guide) is updated.

7.4 Sprint Events Plan

Event	Plan
Sprint Planning	Day 1; 2 hours; Product Owner explains the goal, the team selects stories and creates tasks.
Daily Scrum	Every day; 15 minutes; progress, plan and blockers.
Sprint Review	Day 10; 1 hour; demonstration to stakeholders and feedback.
Sprint Retrospective	Day 10; 45 minutes; what went well, what to improve, and action items for Sprint 2.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	**/25**